

Project Proposal: Mental Health in Technology

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1. Problem Definition and Relevance

Mental Health has become one of the fastest-growing yet underaddressed issues in the technology industry. While tech jobs offer some of the highest salaries and in-demand career paths, they also come with intense pressure, long hours, and a culture that discourages workers from seeking help. This project investigates this through the central question: **What workplace and personal factors predict whether a tech worker will seek mental health treatment, and how does mental health interfere with their work?** To answer this, we will combine two OSMI (Open Sourcing Mental Illness) survey datasets from 2014 and 2016, covering thousands of tech workers.

Strategic Aspects

We approach this from two angles: building a measurable Employer Support Score from workplace features to use as an ML predictor, and running separate models on personal versus workplace feature groups to directly compare which better predicts treatment-seeking.

2. Novelty and Importance

Mental health disorders cost the global economy about \$1 trillion each year in lost productivity (WHO, 2022). In tech, burnout and untreated conditions drive high turnover, lower output, and real harm to workers. By identifying predictors of treatment-seeking, we can suggest steps that technology companies can take to support employees better.

Prior research has looked at workplace mental health in general. Dewa et al. (2014) found that burnout is associated with decreased productivity and increased early age retirement, which highlights the real workplace costs of untreated mental conditions. In software engineering, Russo & Stol (2022) studied gender differences in personality traits amongst software engineers but did not examine treatment-seeking behavior or employer support. The Open Sourcing Mental Illness Survey has also appeared in prior Kaggle projects and in a merged dataset by Memon (2023), combining surveys from 2017–2021. However, that dataset only uses about 8–10 features and focuses on predicting current disorder status. No public analysis merges the 2014 and 2016 surveys with richer features, predicts treatment-seeking and work interference, or compares the predictive impact of personal factors versus workplace factors.

Four Key Contributions

Our project makes four main contributions that set it apart from prior work using this dataset:

1. 2014 + 2016 schema alignment: The two surveys share no matching column names. Our custom alignment produces a unified 2,692-row dataset that no prior analysis has built.
2. Treatment-seeking as the ML target: We predict whether someone has ever sought treatment rather than current disorder status, better capturing help-seeking behavior.
3. Personal vs. workplace feature comparison: Separate models are trained on each feature group, and AUC scores are compared to determine which better predicts treatment-seeking.
4. Two-year comparison: Merging both surveys lets us compare treatment rates, stigma, and employer support across years, adding a temporal dimension.

3. Data Description

We use two OSMI Mental Health in Tech Surveys datasets, both publicly available on Kaggle and accessible for download. Both datasets were created by OSMI and have been downloaded and verified as accessible, requiring no API keys.

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Dataset	Source	Year	Rows	Col
OSMI 2014 Survey	Kaggle	2014	1259	27
OSMI 2016 Survey	Kaggle	2016	1433	63
Combined	Kaggle	2014 & 2016	2692 Rows	89 Col

We evaluated all OSMI surveys from 2014 through 2021. The 2014 and 2016 surveys are the only years with clean, structured versions of all required variables. Years 2017–2021 dropped key variables like anonymity, care_options, and mental_health_consequence, making them unusable for our research question. Post-2021 data has not yet been released.

Known Data Challenges

Column Mismatch: 2016 data represents the columns as questions. Requires manual mapping.

Age: Contains invalid entries, including above 100, negative numbers, and birth years.

Gender: Unique answers that will require standardization into Male, Female, Non-binary, and other

Missing Values: work_interfere nulls will be encoded as Not Applicable instead of dropped. Columns in 2016 with over 80% nulls will be removed.

4. Methodology & Implementation Plan

Data Pipeline

Stage 1 — Data Profiling & Ingestion: Load both CSVs into Pandas DataFrames, generate quality reports (null rates, data types, distributions), store in SQLite, and document schema differences as baseline.

Stage 2 — Schema Alignment: Map 2016 column names to the 2014 equivalents. Record all mapping decisions, including columns that only appear in one year. Add a `survey\_year` column before merging.

Stage 3 — Data Cleaning: Filter age to 18–75, standardize gender into categories, and encode work_interfere nulls as Not Applicable. Drop 2016 columns over 80% null. Merge 2016 treated/untreated work_interfere into a single worst-case measure.

Stage 4 — Merging & Validation: Combine aligned DataFrames and validate by checking row counts, null rates, feature distributions, and target class balance.

Stage 5 — Feature Engineering & Encoding: Build Employer Support and Workplace Stigma composite scores (normalized 0–1), encode target variables, one-hot encode nominal features, and scale age and composite scores with MinMaxScaler.

Machine Learning Models

Model	Role	Justification
Logistic Regression	Baseline model	Coefficient-level interpretability; establishes performance floor; strong for binary classification on survey data
Random Forest	Primary model	Handles mixed data types and non-linear relationships; provides built-in feature importance scores central to our analysis

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Three sub-experiments: (1) Personal vs. workplace features — train Random Forest on each feature group separately and compare ROC-AUC scores. (2) Secondary target — train models using work_interfere as a four-class target. (3) Year as a feature — include survey_year as a binary feature and evaluate its importance ranking.

Evaluation Metrics

Accuracy(Baseline), Precision and Recall(treatment-seeking), F1-Score, ROC-AUC(primary for personal vs workplace experiment), Feature importance (Random Forest), and Confusion Matrix(all classifiers)

Visualizations

World map of treatment-seeking by country; year-over-year bar charts (treatment rates, stigma, employer support); Employer Support Score vs. treatment-seeking scatter plot; Stigma Score vs. treatment avoidance bar chart; comfort discussing mental health by relationship type; top 15 feature importance chart; model performance comparison (F1 and AUC); confusion matrices; personal vs. workplace AUC comparison; age distribution by treatment status.

Tools & Technologies

Tools: Python, Pandas, NumPy, SQLite, scikit-learn (Logistic Regression, Random Forest), Matplotlib, Seaborn, Plotly, Jupyter Notebook, GitHub.

5. Expected Outcomes and Limitations

We expect workplace environment features, such as employer support, to be stronger predictors of treatment-seeking behavior than personal factors such as age and gender. This implies that the technology companies seeking to improve employee mental health should prioritize benefits and wellness programs rather than demographic programs. We additionally expect to find that the years from 2014 to 2016 have measurable predictive value, reflecting changes in the industry.

Limitations

- Self-Report Bias: Respondents may underreport stigmatized behaviors like avoiding treatment.
- Sampling Bias: OSMI reaches mental-health-aware respondents, possibly understating the problem's scale.
- Scope: Both datasets predate COVID-19 and represent a pre-pandemic historical baseline.
- Cross-Sectional Design: These are one-time surveys, so we can find patterns but cannot prove causation.

References

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